

MUZZAMMIL SARWAR

Senior Full-Stack AI Engineer | TypeScript · Python · LLMs & Agents · AI-Assisted Development

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SUMMARY

Senior full-stack AI engineer with 5 years building product platforms, AI systems and data tooling in TypeScript and Python. Currently sole engineer and architect of a production workforce platform (web portals, NestJS API, cross-platform desktop agent, AI features), built quickly with AI-assisted development and gated by tests and code review. Author of pgrecon, an open-source Oracle-to-PostgreSQL migration tool published on PyPI. Open to remote roles.

SKILLS

Languages: TypeScript, Python, JavaScript, SQL, C#, Bash

Backend: Node.js, NestJS, FastAPI, Express, REST APIs, Redis, BullMQ, Webhooks

Frontend: Next.js (App Router), React, Angular, Tailwind CSS

Data: PostgreSQL, pgvector, Prisma, MySQL, Oracle-to-PostgreSQL Migration, Query Optimisation

AI: OpenAI, Gemini Vision, RAG & Embeddings, CrewAI, LangChain, n8n, LightFM

Platform: Electron (Windows, macOS, Linux), Docker, GitHub Actions, Cloudflare, Caddy, Jest

AI-Assisted Development: Claude Code and Cursor, with tests and code review gating every change

EXPERIENCE

Growth Rise Solutions · Remote

Dec 2024 – Present

Full-Stack Developer

May 2026 – Present

Sole engineer and architect of Controla-VA, a workforce time-tracking and productivity platform for distributed teams

- Architected and built the platform end to end: five role-based Next.js portals on a NestJS API (30+ modules, PostgreSQL, Prisma) with office-scoped access control and a full audit trail.
- Built the Electron desktop agent running on ~50 workstations across Asia and Latin America, with Windows and Linux builds (macOS pending Apple signing): offline-first sync, multi-monitor screenshots, native URL capture and silent auto-updates.
- Root-caused an over-billing defect (a 17-minute session recorded as 16.5 hours) and fixed it server-side for every installed version; designed an event-sourced billing model now running in shadow mode.
- Shipped AI features to production: RAG candidate matching on pgvector, Gemini vision screenshot classification, and LLM invoice and CV parsing.
- Built a multi-currency finance platform that reconciles bank, PayPal, Payoneer, Airwallex and eBay statements into a three-tier P&L, with a grounded AI assistant.
- Integrated Zoom meetings with recording and AI summaries; run staging and production on Docker, Caddy and Cloudflare with CI/CD gated by automated tests.

Python / AI Engineer

Dec 2024 – Apr 2026

- Engineered autonomous multi-agent AI systems with CrewAI and LangChain, with custom memory modules and tool orchestration for multi-step tasks.
- Shipped 10+ production automations: event-driven n8n workflows connecting AI agents to CRMs, messaging services and data sources for hands-off business processes.
- Built FastAPI backends serving AI features to hundreds of daily users, plus SEO-friendly Next.js and Streamlit front ends.
- Built a hybrid (collaborative + content-based) LightFM recommendation microservice for a live grocery app (box.com.pk) with thousands of products and shoppers, improving product discovery.
- Containerised services with Docker and automated deployments to Railway through GitHub Actions; owned versioned releases, monitoring and rollback.

Software Engineer | Bitnine Global · Islamabad

Apr 2023 – Dec 2024

- Designed a Python test-automation framework for regression, smoke and sanity testing across Bash, PowerShell and CMD, automating over 70% of manual regression testing.
- Engineered an Oracle-to-PostgreSQL migration tool in Python with both a CLI and a desktop GUI: DDL extraction, schema transformation and end-to-end data migration.
- Architected the migration tool for extensibility so additional databases (DB2, MySQL) plug in without rewriting core logic.

- Built Angular and AngularJS front ends and Express.js REST APIs on MySQL for a full-stack web application.
- Integrated Strapi as a headless CMS to streamline content management and speed up back-end API development.

PROJECTS

pgrecon (open source, github.com/Muzzammil242/pgrecon): Oracle-to-PostgreSQL migration assessment and schema conversion tool, published on PyPI (16 releases, 27 GitHub stars). Inventories an Oracle database from an offline dump, runs a deterministic rule engine that flags the constructs that drive migration cost, and emits PostgreSQL DDL with an explicit refusal for anything it cannot prove. Python, SQLite, CI.

Smart Cold Storage Dashboard (final-year project, NUST): full-stack dashboard for cold-storage automation with real-time temperature monitoring, plus ML models that forecast temperature trends and recommend storage conditions for predictive maintenance and energy efficiency.

Kids Learning Platform (freelance): web platform pairing an eye-tracking model for real-time engagement analysis with APIs around a fine-tuned GPT-3 model that personalises learning content; owned front end and back end.

Case studies: muzzammil242.github.io/#case-studies (architecture write-ups across platform, AI and data work)

EDUCATION

BS Computer Science | National University of Sciences and Technology (NUST), SEECS, Islamabad

2019 – 2023

Coursework: Artificial Intelligence, NLP, Distributed Computing, Digital Image Processing, Database Systems, Operating Systems

Languages: English (professional), Urdu (native)